ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

4C6kBhezHE2GhCECbZRwv9YcH2PNTZ3oEa1pN9Azpump

Date: 5/1/2025
Compounds: Dextromethorphan, PCP

New Compound Analysis

Name: Dexcyphencyl
Formula: $C_{18}H_{25}NO$
Weight: 271.4
Drug-likeness: 0.6

Activity:
NMDA receptor antagonism with cough suppressant properties

Target Analysis

NMDA receptor
Binding Affinity: -8.2
Targeting NMDA receptors is critical for treatments involving anesthesia, neuroprotection, and cough suppression.

Toxicity Profile

Risk Level: high
Affected Systems: brain, liver

Mitigation Suggestions:
- Monitor liver function
- Use in controlled dosages
- Possibly co-administer with neuroprotective compounds

Pharmacokinetics

Absorption:
Rapid gastrointestinal absorption

Distribution:
Widely distributed in tissues, including the brain with a high volume of distribution

Metabolism:
Extensive hepatic metabolism primarily via CYP2D6

Excretion:
Renal excretion of metabolites

Half-life: 3-6 hours

Detailed Analysis

Dexcyphencyl, a hybrid of Dextromethorphan and PCP, leverages properties of both compounds to engage the NMDA receptor in a non-competitive manner. This proposed compound is synthesized to exhibit stronger NMDA antagonism with secondary cough-suppressing capabilities from Dextromethorphan while aiming to mitigate PCP's psychotomimetic effects. Its pharmacokinetic profile indicates rapid absorption and extensive distribution that warrants close monitoring due to potential neurotoxicity and liver impact. Although presenting promising clinical applications in cough suppression and anesthesia, its development requires strategies to address drug-likeness factors and mitigate psychotropic risks.
